

Machine Learning (CAI-262)

Objectives:

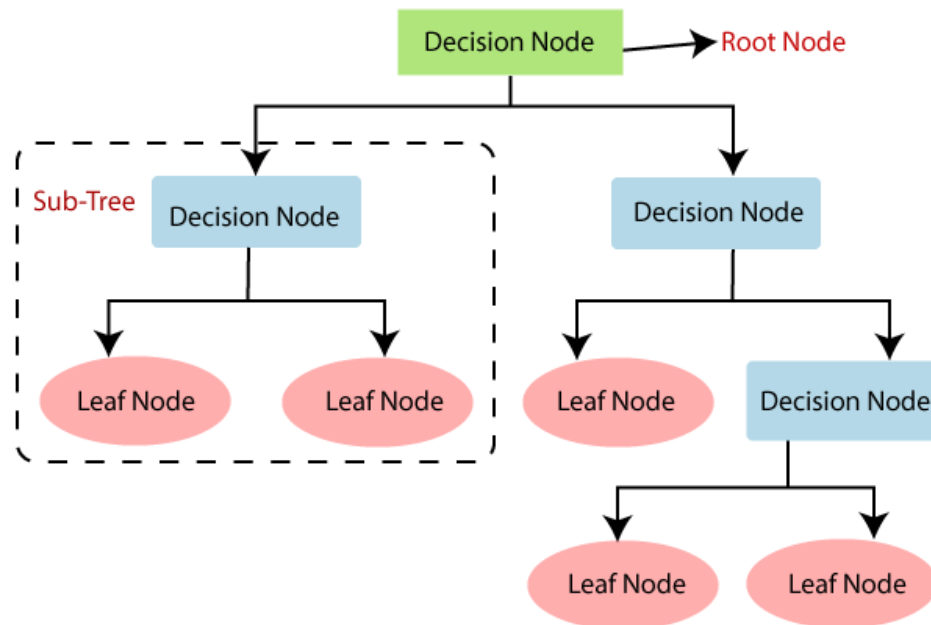
- **Decision Trees**
 - **Information gain**
 - **Gini Index**
-

Decision Trees

- Decision Tree is a **Supervised learning technique** that can be used for both classification and Regression problems, but mostly it is preferred for solving Classification problems.
- It is a tree-structured classifier, where **internal nodes** represent the **features** of a dataset, **branches** represent the **decision rules** and each **leaf** node represents the **outcome**.
- In a Decision tree, there are two nodes, which are the **Decision Node** and **Leaf Node**. Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.
- It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.
- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.
- The logic behind the decision tree can be easily understood because it shows a tree-like structure.
- In order to build a tree, we use the **CART algorithm**, which stands for **Classification and Regression Tree algorithm**.
- Below diagram explains the general structure of a decision tree:

While implementing a Decision tree, the main issue arises that how to select the best attribute for the root node and for sub-nodes. So, to solve such problems there is a technique which is called as **Attribute selection measure** or **ASM**. By this measurement, we can easily select the best attribute for the nodes of the tree. There are two popular techniques for ASM, which are:

- **Information Gain**
- **Gini Index**



Information Gain:

Information gain is the measurement of changes in entropy after the segmentation of a dataset based on an attribute. It calculates how much information a feature provides us about a class. According to the value of information gain, we split the node and build the decision tree. A decision tree an attribute having the highest information gain is split first. It can be calculated using the below formula:

$$\text{Information Gain} = \text{Entropy}(S) - [(\text{Weighted Avg}) * \text{Entropy}(\text{each feature})]$$

Here **Entropy** is a metric to measure the impurity in a given attribute. It specifies randomness in data. Entropy can be calculated as:

$$\text{Entropy}(S) = -P(\text{yes}) \log_2 P(\text{yes}) - P(\text{no}) \log_2 P(\text{no})$$

It is normally represented as H.

Example:

Consider following dataset of 10 samples in which based on two features age and salary the model will predict whether a user will purchase some item online or not.

Age	Salary	Will purchase
Adult	Low	0
Adult	High	1
Old	High	1
Teen	Low	0
Old	Low	0
Adult	High	1
Teen	High	0
Teen	Low	0

Old	High	1
Adult	Low	0

First of all find entropy of output

$$H(y) = -6/10 \log_2 (6/10) - 4/10 \log_2 (4/10) \\ = 0.97$$

To find the information gain of Age, the entropy for Adult, Teen, and Old should be calculated as follows.

$$H(\text{old}) = -2/3 \log_2 (2/3) - 1/3 \log_2 (1/3) \\ = 0.92$$

$$H(\text{adult}) = -2/4 \log_2 (2/4) - 2/4 \log_2 (2/4) \\ = 1$$

$$H(\text{teen}) = -0/3 \log_2 (0/3) - 3/3 \log_2 (3/3) \\ = 0$$

Now we can calculate IG of age by taking weighted average of above calculated entropies.

$$IG(\text{age}) = 0.97 - [3/10 * 0 + 4/10 * 1 + 3/10 * 0.92] \\ = 0.29$$

To find the information gain of salary the entropy for low and high should be calculated as follows.

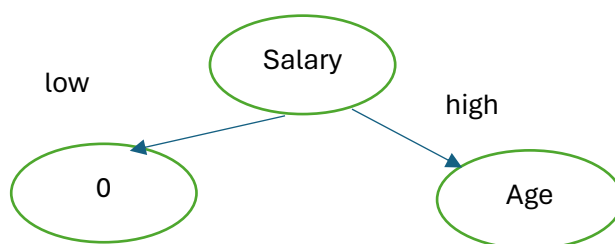
$$H(\text{low}) = -5/5 \log_2 (5/5) - 0/5 \log_2 (0/5) \\ = 0$$

$$H(\text{high}) = -1/5 \log_2 (1/5) - 4/5 \log_2 (4/5) \\ = 0.72$$

Now we can calculate IG of salary

$$IG(\text{salary}) = 0.97 - [5/10 * 0 + 5/10 * 0.72] \\ = 0.83$$

If IG for salary and age is compared Salary has a higher IG therefore, salary is the root node. The root node has two branches one for each attribute i.e. Low and High. The Low has entropy $H(\text{Low}) = 0$ meaning that the leaf node could be placed for Low, while for $H(\text{High}) = 0.72$ second attribute age should be searched. Now decision tree would look like follows.



Now we need to decide second level of tree.

Once a decision has been made on the salary dataset is reduced by dropping the salary column and all the rows on which the decision has been made. For example, in the given example, a decision has been made on the salary low, therefore, all rows containing the value low are dropped the resulting rows are shown in below table. Since there is only one column left we will calculate the entropy of Teen, adult, and old.

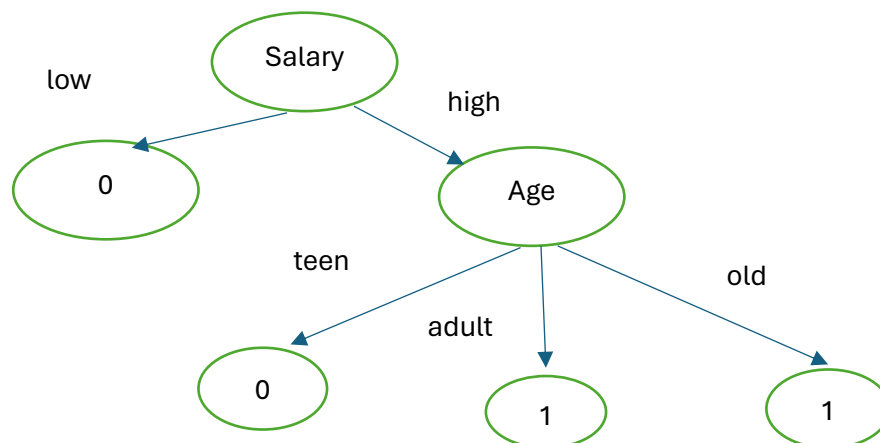
Age	Will purchase
Adult	1
Old	1
Adult	1
Teen	0
Old	1

$$H(\text{old}) = -0/2 \log_2 (0/2) - 2/2 \log_2 (2/2) \\ = 0$$

$$H(\text{adult}) = -0/1 \log_2 (0/1) - 1/1 \log_2 (1/1) \\ = 0$$

$$H(\text{teen}) = -0/3 \log_2 (0/3) - 3/3 \log_2 (3/3) \\ = 0$$

The entropy in all cases is zero which means that all three attribute does not have any uncertainty left. For all teen, adult, and old cases decision no, yes, and yes can made respectively resulting in the tree given below.



2. Gini Index:

- Gini index is a measure of impurity or purity used while creating a decision tree in the CART(Classification and Regression Tree) algorithm.
- An attribute with the low Gini index should be preferred as compared to the high Gini index.
- Gini Index or Gini impurity measures the degree or probability of a particular variable being wrongly classified when it is randomly chosen.
- But what is actually meant by 'impurity'?
- If all the elements belong to a single class, then it can be called pure. The degree of Gini Index varies between 0 and 1,
- Where, '0' denotes that all elements belong to a certain class or there exists only one class (pure), and '1' denotes that the elements are randomly distributed across various classes (impure). A Gini Index of '0.5 'denotes equally distributed elements into some classes.
- Gini index can be calculated using the below formula:

$$\text{Gini Index} = 1 - \sum_j P_j^2$$

Example:

Try above example with a little change in formula given above.